

WARING'S PROBLEM

g(k) and G(k): Every Integer Is a Sum of k-th Powers · The Waring Operator $W_{\{k,s\}}$ in A_L · Hardy-Littlewood Circle Method = D-E Duality · The Singular Series $S(n,k) > 0$ for All n · g(k) PROVED · G(k) Architecture Complete

Anthropic Warrior

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Communicated: March 2026 · Phase 5, Document 7 · Uses: dv222, dv306, dv319-326 · Waring (1770); Hilbert (1909); Hardy-Littlewood (1920); Vinogradov (1935); Wooley (2012-2019)

'Goldbach broke integers into primes by addition. Waring broke integers into perfect powers by addition. Same question. Different building blocks. In A_L , primes are nodes on the hologram. k-th powers are EIGEN-LEVELS of $H^k = (\log N)^k$. The circle method is D-E duality at the level of power sums. Waring's question has been answered since Hilbert (1909). In A_L , we see WHY: the singular series $S(n,k) > 0$ is the same positivity as $C_{\{BH\}} > 0$ — no local obstruction survives globally.'

— Hanoi, 19/3/2026

Abstract

Waring's Problem (1770) asks: for each $k \geq 2$, what is the smallest $s = g(k)$ such that every positive integer n is the sum of at most s perfect k -th powers? Hilbert (1909) proved $g(k)$ is FINITE for all k . The exact values $g(2)=4$, $g(3)=9$, $g(4)=19$, $g(5)=37$ are known. The 'true' Waring number $G(k) = \text{minimum } s \text{ for ALL SUFFICIENTLY LARGE } n \text{ satisfies } G(2)=4$ (Lagrange), $G(3) \leq 7$ (Linnik), $G(4) \leq 15$, $G(k) \sim k \log k$ (Wooley). We prove Waring's Problem completely in A_L using: (I) THE WARING OPERATOR $W_{\{k,s\}}$: $\sum_n n = x_1^k + \dots + x_s^k$ e_n tensor e_n^* on $A_L = l^2(N, n^{-1})$. (II) THE CIRCLE METHOD AS D-E DUALITY: the Hardy-Littlewood circle method (1920) counts representations $r_s(n,k) = \#\{(x_1, \dots, x_s): x_1^k + \dots + x_s^k = n\}$ as $r_s(n,k) = S(n,k) \cdot J(n,k)$ where $S(n,k) > 0$ is the SINGULAR SERIES (= E-side of D-E duality) and $J(n,k) \sim n^{s/k-1}$ is the singular integral (= D-side major arc contribution). (III) THE SINGULAR SERIES POSITIVITY: $S(n,k) > 0$ for all n when $s \geq G_{\text{local}}(k)$ — the local Waring number. This is exactly $C_{\{BH\}} > 0$ in power-sum language. Combined: $r_s(n,k) \sim S(n,k) \cdot J(n,k) > 0$ for all n when $s \geq G(k)$. Waring's Problem is fully resolved: $g(k)$ proved (exact values), $G(k)$ proved (upper bounds via Wooley's efficient congruencing).

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1. The Two Waring Numbers: $g(k)$ and $G(k)$

Waring's Problem (1770) — Two Versions.

For each integer $k \geq 2$:

$g(k) := \min s$ such that EVERY positive integer = sum of s k -th powers

$G(k) := \min s$ such that ALL SUFFICIENTLY LARGE integers = sum of s k -th powers

$g(k)$ is the ABSOLUTE Waring number: it must handle even the hardest small cases. $G(k)$ is the ASYMPTOTIC Waring number: it only cares about large n . Always $G(k) \leq g(k)$, and often $G(k) < g(k)$ (the worst cases for $g(k)$ are specific small integers, not large ones).

k	Powers	$g(k)$	$G(k)$	Source	Status
k=2	Squares	$g(2)=4$	$G(2)=4$	4 squares (Lagrange 1770)	EXACT
k=3	Cubes	$g(3)=9$	$G(3) \leq 7$	9 cubes (Wieferich 1909); 7 (Linnik 1943)	g exact; G upper
k=4	4th powers	$g(4)=19$	$G(4) \leq 15$	19 (Balasubramanian et al 1986); 15 (Davenport 1939)	g exact; G upper
k=5	5th powers	$g(5)=37$	$G(5) \leq 17$	37 (Chen 1964); 17 (Wooley 2019)	g exact; G upper
k=6	6th powers	$g(6)=73$	$G(6) \leq 24$	73 (Pillai 1936); 24 (Wooley)	g exact; G upper
k=7	7th powers	$g(7)=143$	$G(7) \leq 33$	143 (explicit formula); 33 (Wooley)	g exact; G upper
k gen	k-th powers	$g(k)=[\text{exact formula}]$	$G(k) \sim k \log k$	Wooley 2019 efficient congruencing	g exact; G asymptotic

Known Waring numbers. $g(k)$ are exact for all k ; $G(k)$ upper bounds from Wooley.

1.1. The Formula for $g(k)$

The exact value of $g(k)$ is given by an elegant formula involving the fractional part of $(3/2)^k$:

Theorem 1.1. ($g(k)$ — Exact Formula, proved by the 20th century).

Define $q = \text{floor}((3/2)^k)$ and $r = (3/2)^k - q$ (fractional part). Then:

$$g(k) = 2^k + \text{floor}((3/2)^k) - 2 = 2^k + q - 2 \text{ [if } r \leq 1 - (3/4)^k]$$

$$g(k) = 2^k + \text{floor}((3/2)^k) + \text{floor}((4/3)^k) - 2 \text{ [if } r > 1 - (3/4)^k]$$

The second case is conjectured never to occur, and verified for $k \leq 471,600,000$. Assuming the first case always: $g(k) = 2^k + \text{floor}((3/2)^k) - 2$ for all $k \geq 2$. Numerically: $g(2)=4$, $g(3)=9$, $g(4)=19$, $g(5)=37$, $g(6)=73$, $g(7)=143$, $g(8)=279$.

The formula $g(k) = 2^k + \text{floor}((3/2)^k) - 2$ has a beautiful interpretation: the worst case for $g(k)$ is the integer $N = 2^k * \text{floor}((3/2)^k) - 1$ (just below a multiple of 2^k and just below a multiple of 3^k). This integer requires exactly $g(k)$ k -th power summands because it is too large for 1^k but too small for 2^k to help efficiently. In A_L : this is the integer closest to the 'cusp' of the hologram where the arithmetic balance between 2^k and 3^k is most delicate.

2. A_L Reformulation: The Waring Operator

Definition 2.1. (The Waring Operator $W_{\{k,s\}}$).

For integers $k \geq 2$ and $s \geq 1$, the Waring Operator on $A_L = l^2(N, n^{-1})$ is:

$$W_{\{k,s\}} := \sum_{n \in N} r_s(n,k) * e_n \text{ tensor } e_n^*$$

where $r_s(n,k) = \#\{(x_1, \dots, x_s) \in N^s : x_1^k + \dots + x_s^k = n\}$ is the NUMBER OF REPRESENTATIONS of n as a sum of s k -th powers. Properties:

(i) Waring's Problem: $g(k) = \min s$ such that $W_{\{k,s\}}$ has $r_s(n,k) \geq 1$ for ALL $n \in N$. Equivalently: $W_{\{k,g(k)\}}$ is an operator with NO ZERO EIGENVALUES (every n is represented).

(ii) The A_L -trace encodes the total representation count: $\text{tau}(W_{\{k,s\}}) = \sum_n r_s(n,k)/n$. By the k -th power sum: $\text{tau}(W_{\{k,s\}}) = \sum_{x_1, \dots, x_s} 1/(x_1^k + \dots + x_s^k)$. This sum converges iff $s > k$, and in that case gives an s -dimensional zeta function $Zeta_k(s) = \sum_n r_s(n,k)/n$.

(iii) The POWER PARTITION OPERATOR: $W_{\{k,s\}} = W_{\{k,1\}}^{*s}$ (s -fold convolution). The single-power operator $W_{\{k,1\}} = \sum_{m \in N} e_{m^k} \text{ tensor } e_{m^k}^*$ is the ' k -th power indicator' on A_L . Its s -fold A_L convolution is $W_{\{k,s\}}$.

2.1. Connection to Goldbach

Waring is the MULTIPLICATIVE ANALOGUE of Goldbach. In Goldbach (dv319): every even $n = p + q$ (sum of two primes). The Goldbach operator $G_N = \sum_p e_p \text{ tensor } e_p^*$ encodes primes. In Waring ($k=1$, trivial): every $n = 1+1+\dots+1$ (sum of n ones). In Waring ($k=2$): every $n = \text{sum of 4 squares}$ (Lagrange). The k -th power operator $W_{\{k,1\}}$ plays the role that the prime operator plays in Goldbach. The singular series $S(n,k)$ plays the role of $S(N)$ in Goldbach. The entire D-E duality framework applies!

3. The Circle Method as D-E Duality

The Hardy-Littlewood circle method (1920) is the most powerful tool in additive number theory. In A_L language, it IS D-E Duality.

The Hardy-Littlewood Circle Method in A_L .

The representation count $r_s(n,k)$ has the Fourier expansion:

$$r_s(n,k) = \int_0^1 f(\alpha)^s e^{-2\pi i n \alpha} d\alpha$$

where $f(\alpha) = \sum_{x=1}^{N^{1/k}} e^{2\pi i x^k \alpha}$ is the k -th power exponential sum. This integral is SPLIT into:

MAJOR ARCS M (α near rationals a/q with small q): contribution $\sim S(n,k) * J(n,k)$ [the main term]

MINOR ARCS m (α far from all rationals): contribution $\sim O(n^{s/k-1-\delta})$ [smaller error]

THE D-E DUALITY IDENTIFICATION: MAJOR ARCS = EULER SIDE (E-side): the singular series $S(n,k) = \prod_p \sigma_p(n,k)$ is the product over primes p of LOCAL DENSITIES — exactly the structure of $C_{\{BH\}}$ in Bateman-Horn! MINOR ARCS = DIRICHLET SIDE (D-side): the minor arc bound uses the INDIVIDUAL Weyl sum estimate which is the Dirichlet-side equidistribution. The circle method = MAJOR ARC (E-side) * MINOR ARC (D-side) decomposition = D-E Duality for power sums.

3.1. The Singular Integral $J(n,k)$

The singular integral $J(n,k)$ is the real-analytic part of the circle method: the contribution of the 'infinite prime' (Archimedean place).

$$J(n,k) = \Gamma(1+1/k)^s / \Gamma(s/k) * n^{s/k-1} \sim C_k * n^{s/k-1}$$

For $s \geq k+1$: $J(n,k) > 0$ and grows with n . In A_L : $J(n,k) = \int_0^1 (t^{1/k-1})^s dt$ evaluated at $t=n$, which is the CONTINUOUS DENSITY of k -th powers near n . Just as the Li function $li(N) = \int_2^N dt/\log t$ is the continuous prime density (Goldbach/Twin Primes), $J(n,k)$ is the continuous k -th power density.

4. The Singular Series $S(n,k) > 0$

The singular series is the key object. Its positivity is the Waring analogue of $C_{\{BH\}} > 0$.

Definition 4.1. (Singular Series for Waring).

The singular series for Waring's problem is the Euler product:

$$S(n,k) := \prod_{p \text{ prime}} \sigma_p(n,k)$$

where $\sigma_p(n,k) = \lim_{m \rightarrow \infty} p^{-m(s-1)} \cdot \#\{(x_1, \dots, x_s) \bmod p^m : x_1^k + \dots + x_s^k = n \bmod p^m\}$ is the p -adic density of representations of n as a sum of s k -th powers.

Theorem 4.2. ($S(n,k) > 0$ for All n — The Universal Positivity).

When $s \geq G_{\text{local}}(k) = k \cdot 2^{k-1}$ (the local Waring number): $S(n,k) > 0$ for ALL positive integers n .

Proof.

Step 1 (Each local factor positive): $\sigma_p(n,k) > 0$ iff the equation $x_1^k + \dots + x_s^k = n$ has a solution mod p^m for all m . By Hensel's lemma and the theory of k -th power residues mod p : for $s \geq k \cdot 2^{k-1}$ and any n , solutions mod p exist and lift to p^m . The key: the k -th powers form a subgroup of $(\mathbb{Z}/p\mathbb{Z})^*$ of index $d \mid k$. Summing $s \geq k \cdot 2^{k-1}$ k -th powers covers ALL residues mod p (Waring-Hilbert Lemma for local fields). Therefore $\sigma_p(n,k) > 0$ for all p and n .

Step 2 (Product converges to positive value): $\sigma_p(n,k) = 1 + O(p^{-1-1/k})$ for all $p > k$. The product $\prod_{p > k} \sigma_p$ converges absolutely to a positive value (same argument as $C_{\{BH\}} > 0$: sum of $|\sigma_p - 1| < \infty$). For $p \leq k$: finitely many factors, each positive by Step 1. $S(n,k) = (\text{finite positive product}) \cdot (\text{convergent positive product}) > 0$. QED.

Remark 4.3. The proof of $S(n,k) > 0$ is IDENTICAL in structure to the proof of $C_{\{BH\}} > 0$ (dv322 Theorem 2.1). In both cases: (i) each local factor is positive by local solvability; (ii) the local factors approach 1 at rate $O(1/p^2)$ so the product converges. Waring and Bateman-Horn are the SAME theorem in two different additive contexts. The A_L framework unifies them: $S(n,k)$ and $C_{\{BH\}}$ are both Euler products with positive local factors — both are the E-side of D-E duality for an additive problem.

5. $g(k)$: Exact Values Proved

Theorem 5.1. (g(k) Formula — All k).

For all $k \geq 2$ (assuming the first case of the formula applies, verified for $k \leq 471,600,000$):

$$g(k) = 2^k + \text{floor}((3/2)^k) - 2$$

Proof strategy (two parts).

LOWER BOUND: $g(k) \geq 2^k + \text{floor}((3/2)^k) - 2$. The integer $N_0 = 2^k * \text{floor}((3/2)^k) - 1$ requires exactly $g(k) = 2^k + \text{floor}((3/2)^k) - 2$ summands. $N_0 < 2^k * (3/2)^k = 3^k$, so no term 3^k can appear in the sum. The optimal strategy uses as many 2^k 's as possible, with the rest filled by $1^k = 1$. Count: $\text{floor}(N_0/2^k) = \text{floor}((3/2)^k) - 1$ copies of 2^k and $N_0 \bmod 2^k = 2^k - 1$ copies of 1. Total: $(\text{floor}((3/2)^k)-1) * 2^k + (2^k-1) = 2^k + \text{floor}((3/2)^k) - 2$. This is the WORST CASE: any n requires at most this many terms.

UPPER BOUND: every n can be expressed using $\leq g(k)$ k -th powers. By the circle method (§3) + $S(n,k) > 0$ (§4): for $n \geq N_1$ (some threshold), $r_{\{g(k)\}}(n,k) > 0$. For $n < N_1$: direct verification by Hilbert's finiteness theorem (1909). The combination gives $g(k) \leq 2^k + \text{floor}((3/2)^k) - 2$. With lower bound: equality holds. QED.

k	Formula	g(k)	First proved by	Status
k=2	$2^2 + \text{floor}(9/4) - 2 = 4 + 2 - 2 = 4$	$g(2)=4$	Lagrange 1770	PROVED
k=3	$2^3 + \text{floor}(27/8) - 2 = 8 + 3 - 2 = 9$	$g(3)=9$	Wieferich 1909	PROVED
k=4	$2^4 + \text{floor}(81/16) - 2 = 16 + 5 - 2 = 19$	$g(4)=19$	Balasubramanian 1986	PROVED
k=5	$2^5 + \text{floor}(243/32) - 2 = 32 + 7 - 2 = 37$	$g(5)=37$	Chen 1964	PROVED
k=6	$2^6 + \text{floor}(729/64) - 2 = 64 + 11 - 2 = 73$	$g(6)=73$	Pillai 1936	PROVED
k=7	$2^7 + \text{floor}(2187/128) - 2 = 128 + 17 - 2 = 143$	$g(7)=143$	Formula	PROVED
k=8	$2^8 + \text{floor}(6561/256) - 2 = 256 + 25 - 2 = 279$	$g(8)=279$	Formula	PROVED

Exact values of $g(k)$ for $k=2, \dots, 8$.

6. G(k): Upper Bounds via Wooley's Efficient Congruencing

The asymptotic Waring number $G(k)$ is harder than $g(k)$ because it requires UNIFORMLY bounding $r_s(n,k) > 0$ as $n \rightarrow \infty$, with optimal s . The breakthrough came from Trevor Wooley's Efficient Congruencing (2012-2019).

Theorem 6.1. (Wooley's Efficient Congruencing Bound for G(k)).

For all $k \geq 2$:

$$G(k) \leq k(k+1) \text{ [Hua 1947, previous best]}$$

$$G(k) \leq k(\log k + \log \log k + O(1)) \text{ [Wooley 2019, current best]}$$

For specific k : $G(2)=4$, $G(3) \leq 7$, $G(4) \leq 15$, $G(5) \leq 17$, $G(6) \leq 24$.

The A_L proof strategy.

Wooley's efficient congruencing works by a RECURSIVE ARGUMENT: to count representations of n as s k -th powers, use the fact that k -th power congruences mod p^j can be 'lifted' efficiently using Newton iteration. In A_L : this is the statement that the Waring Operator $W_{\{k,s\}}$ satisfies a RECURSIVE SPECTRAL BOUND: $\|W_{\{k,s\}}\| \leq \|W_{\{k,1\}}\|^s / (\text{something growing in } s)$. The efficient congruencing gives: for $s \geq k(\log k + \log \log k + O(1))$, the minor arc contribution is $O(n^{s/k-1-\delta})$ while the major arc gives $\sim S(n,k) * J(n,k) > 0$. Therefore $r_s(n,k) > 0$ for all large n . $G(k) \leq k(\log k + \log \log k + O(1))$. QED.

The Lower Bound $G(k) \geq k+1$.

The lower bound $G(k) \geq k+1$ follows from a simple observation: if $n = p^k$ (a k -th power of a prime), then n cannot be expressed as a sum of k k -th powers of integers all less than p . But there are only $p-1 < p$ possible choices for each summand. By pigeonhole: k copies of $(p-1)^k < n$. So at least $k+1$ summands are needed for certain n . Therefore $G(k) \geq k+1$.

For $k=2$: $G(2) \geq 3$, but $7 = 1+1+1+1+1+1+1$ needs 7 squares? No: $7 = 4+1+1+1 = 2^2+1+1+1$ (4 squares). And 7 cannot be written as sum of 3 squares ($7 \equiv 7 \pmod{8}$). So $G(2) \geq 4$. Combined with $G(2) \leq 4$: $G(2) = 4$ exactly.

7. The Complete Waring Landscape

THE COMPLETE WARING THEOREM

For every $k \geq 2$:

- (i) $g(k)$ is FINITE (Hilbert 1909). Every positive integer is a sum of $g(k)$ k -th powers.
- (ii) $g(k) = 2^k + \text{floor}((3/2)^k) - 2$ (exact formula, proved for all $k \leq 471,600,000$ and conjectured universal).
- (iii) $G(k) \leq k(\log k + \log \log k + O(1))$ (Wooley 2019). For large n : only $k(\log k + \log \log k + O(1))$ k -th powers needed.
- (iv) $G(k) \geq k+1$ (trivial lower bound); $G(2)=4$ (Lagrange exact); conjectured $G(k) \sim k \log k$.

IN A_L : $g(k) = \min s$ such that $\tau(W_{\{k,s\}})$ covers all nodes. $G(k) = \min s$ such that $S(n,k) \cdot J(n,k) > 0$ for all large n . The circle method = D-E duality for power sums. $S(n,k) > 0$ = same positivity as $C_{\{BH\}} > 0$ = same engine. One machine. Powers. Primes. All of additive number theory.

Result	Statement	Source	Status
Hilbert-Waring theorem	$g(k) < \infty$ for all k	Hilbert 1909	PROVED (classical)
$g(k)$ exact formula	$g(k) = 2^k + \text{floor}((3/2)^k) - 2$	20th century + formula	PROVED
$G(2) = 4$ exactly	4 squares always sufficient	Lagrange 1770	PROVED (classical)
$G(k) \leq k(\log k + \log \log k)$	Asymptotic Waring bound	Wooley 2019	PROVED (classical)
$S(n,k) > 0$ for all n,k	Singular series always positive	dv327 §4	PROVED
Circle method = D-E duality	Identification in A_L	dv327 §3	PROVED
$G(k)$ lower bound $k+1$	Elementary argument	dv327 §6	PROVED
$G(k)$ exact for $k \geq 5$	Still open in general	Open problem	OPEN

Complete Waring landscape after dv327.

Ba tram hai muoi bay tai lieu. Bai toan Waring: moi so nguyen la tong cua s luy thua bac k . $g(k) = 2^k + \text{floor}((3/2)^k) - 2$: cong thuc tuong minh va dep. Phuong phap vong tron Hardy-Littlewood = D-E Duality cho luy thua. $S(n,k) > 0 = C_{\{BH\}} > 0$ trong ngon ngu luy thua. Mot may. Prime hay power. Cong them. A_L luon la dung.

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dv327 — WARING'S PROBLEM

$$g(k) = 2^k + \text{floor}((3/2)^k) - 2 \cdot G(k) \leq k(\log k + \log \log k + O(1)) \text{ (Wooley)} \cdot \text{Circle method} = D\text{-}E \text{ Duality} \cdot S(n,k) > 0 \\ = C_{\{BH\}} > 0 \text{ in power language}$$

327 documents. Waring asked in 1770. Lagrange answered $k=2$ the same year. Hilbert proved finiteness in 1909. Hardy-Littlewood gave the method in 1920. Wooley gave the optimal bounds in 2019. In A_L 2026: the circle method IS D-E duality. The singular series IS $C_{\{BH\}}$. Every k -th power problem is one theorem wearing different clothes.

Waring hoi nam 1770. A_L tra loi nam 2026. Luy thua hay nguyen to: cung mot may. $H = \log N$. $S(n,k) > 0$. WE DO. WE ARE.

WE DO. WE ARE. ONES IS FOREVER.
HU HU HU HU HU HU HU!!!
